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Los de volgende logaritmische vergelijking op:

$$\log_{x+3} x \cdot \log_{10} x + \log_{10} x - \log_{x+3} x = 0$$

$$\Leftrightarrow \log_{x+3} x \cdot \log_{10} x = \log_{x+3} x - \log_{10} x$$

$$\Leftrightarrow \log_{10} x = \frac{\log_{x+3} x - \log_{10} x}{\log_{x+3} x} = 1 - \frac{\log_{10} x}{\log_{x+3} x} = 1 - \frac{\log_{10} x}{\frac{\log_{10} x}{\log_{10}(x+3)}}$$

Geval 1: $x \neq 1$

$$\log_{10} x = 1 - \frac{\cancel{\log_{10} x}}{\frac{\cancel{\log_{10} x}}{\log_{10}(x+3)}} = 1 - \log_{10}(x+3)$$

$$\Leftrightarrow \log_{10}(x \cdot (x+3)) = 1$$

$$\Leftrightarrow 10 = x^2 + 3x$$

$$\Leftrightarrow x^2 + x - 10 = 0$$

$$D = 49 = 7^2$$

$$x_1 = \frac{-3-7}{2} = \cancel{5}$$

$$x_2 = \frac{-3+7}{2} = 2$$

Geval 2: $x = 1$

$$\log_{x+3} 1 \cdot \log_{10} 1 + \log_{10} 1 - \log_{x+3} 1 = 0$$

$$V = \{2, 1\}$$

References